

1 Higher-order linear ODE theory



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

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Lecture 15
Higher-order linear ODE theory
Friday 9 October 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

n^{th} order linear IVPs

The most general n^{th} **order linear ODE** can be written

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (1)$$

where the $n+2$ functions $a_n(x)$, $a_{n-1}(x)$, ..., $a_0(x)$, $g(x)$, are defined for all x in an interval $\mathcal{I}$.

- The functions $a_i(x)$, for $i = 0, 1, \dots, n$, and $g(x)$ *can be nonlinear functions of x* .
- If $g(x) \equiv 0$ then the ODE is **homogeneous**.
- The linear ODE (1) is **normal on $\mathcal{I}$** if $a_0(x), \dots, a_n(x)$ and $g(x)$ are continuous on $\mathcal{I}$, and $a_n(x) \neq 0$ throughout $\mathcal{I}$.

We obtain an n^{th} **order Initial Value Problem (IVP)** if we specify n initial conditions:

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}, \quad (2)$$

where the n numbers $y_0, y_1, \dots, y_{n-1} \in \mathbb{R}$ are constants.

n^{th} order linear IVPs

Theorem (Existence and Uniqueness for Linear ODEs [Trench 9.1.1])

Let $\mathcal{I} = (a, b)$ be an interval and $x_0 \in \mathcal{I}$. Suppose the ODE (1) is *normal on $\mathcal{I}$* . Then the IVP consisting of the ODE (1) and the initial conditions (2) has a unique solution $y(x)$ defined on *all* of $\mathcal{I}$.

Note: Unlike the existence-uniqueness theorem that we discussed for first order ODEs, the theorem above guarantees a unique solution on the *whole interval $\mathcal{I}$* . But this theorem applies only to *linear* ODEs.

Differential Operators

Derivative operator D :

$$Dy = \frac{dy}{dx}, \quad D^2y = D(Dy) = \frac{d^2y}{dx^2}, \quad \dots, \quad D^n y = \frac{d^n y}{dx^n}$$

n^{th} order linear differential operator:

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \dots + a_1(x)D + a_0(x)$$

Any such L is linear: If $y_1(x)$ and $y_2(x)$ are n -times differentiable functions and $\alpha_1, \alpha_2 \in \mathbb{R}$ then:

$$L[\alpha_1 y_1(x) + \alpha_2 y_2(x)] = \alpha_1 L[y_1(x)] + \alpha_2 L[y_2(x)]$$

The most general n^{th} order linear ODE (1) can be written

$$L[y] = g(x)$$

Superposition Principle

When is the sum of solutions to an ODE also a solution?

Answer: Whenever the ODE is *linear* and *homogeneous*.

Theorem (Superposition Principle [Trench §9.1])

If $y_1, y_2, \dots, y_k$ are solutions of $L[y] = 0$ on an interval $\mathcal{I}$, then any linear combination

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x),$$

is also a solution of $L[y] = 0$ on the interval $\mathcal{I}$.

Proof on next slide...

Note: The terminology “superposition” comes from wave theory.

Superposition Principle

Proof of Superposition Principle.

Suppose $L[y_i] = 0$ for each $i = 1, 2, \dots, k$.

$$\begin{aligned} & L[c_1 y_1 + c_2 y_2 + \cdots + c_k y_k] \\ &= L[c_1 y_1] + L[c_2 y_2 + \cdots + c_k y_k] && L \text{ is linear} \\ &= L[c_1 y_1] + L[c_2 y_2] + L[c_3 y_3 + \cdots + c_k y_k] && L \text{ is linear} \\ &\vdots \\ &= L[c_1 y_1] + L[c_2 y_2] + \cdots + L[c_k y_k] && L \text{ is linear} \\ &= c_1 L[y_1] + c_2 L[y_2] + \cdots + c_k L[y_k] && L \text{ is linear} \\ &= c_1 \cdot 0 + c_2 \cdot 0 + \cdots + c_k \cdot 0 && L[y_i] = 0 \\ &= 0 \end{aligned}$$

$\Rightarrow c_1 y_1 + c_2 y_2 + \cdots + c_k y_k$ is a solution of $L[y] = 0$.



Superposition Principle

Important special cases of the superposition principle:

- For a homogeneous linear ODE, a constant multiple of a solution is a solution: $L[cy] = c L[y] = c \cdot 0 = 0$ for any $c \in \mathbb{R}$
- Considering $c = 0$: the trivial solution, $y(x) \equiv 0$, is always a solution to a homogeneous ODE: $L[0] = 0$.

Linear independence and dependence of functions

Definition (Linear independence/dependence)

The k functions $f_1(x), \dots, f_k(x)$ defined on the interval $\mathcal{I}$ are said to be **linearly independent on $\mathcal{I}$** if whenever their linear combination vanishes on $\mathcal{I}$, i.e.,

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_k f_k(x) = 0 \quad \text{for all } x \in \mathcal{I}$$

it follows that $c_1 = c_2 = \cdots = c_k = 0$.

- If $f_1(x), \dots, f_k(x)$ are not linearly independent on $\mathcal{I}$, they are said to be **linearly dependent on $\mathcal{I}$** .
- If they are linearly dependent, then at least one of the functions $f_1(x), \dots, f_k(x)$ can be expressed as a linear combination of the others (on $\mathcal{I}$).

Linear independence and dependence of functions

Example (Linear independence/dependence)

Show that the functions $\cos x$, $\sin x$, and the constant function 1, are linearly **independent** on $\mathcal{I} = (-\pi, \pi)$.

Solution: Suppose

$$c_1 \cos x + c_2 \sin x + c_3(1) = 0 \quad \text{for all } x \in (-\pi, \pi).$$

For $x = 0$ we get:

$$c_1 \cos 0 + c_2 \sin 0 + c_3(1) = c_1 \cdot 1 + c_2 \cdot 0 + c_3 = c_1 + c_3 = 0 \implies c_1 = -c_3$$

whereas for $x = \frac{\pi}{2}$ we get:

$$c_1 \cos \frac{\pi}{2} + c_2 \sin \frac{\pi}{2} + c_3(1) = c_1 \cdot 0 + c_2 \cdot 1 + c_3 = c_2 + c_3 = 0 \implies c_2 = -c_3$$

...cont...

Linear independence and dependence of functions

Example (Linear independence/dependence)

and for $x = -\frac{\pi}{2}$ we get:

$$\begin{aligned} c_1 \cos -\frac{\pi}{2} + c_2 \sin -\frac{\pi}{2} + c_3(1) &= c_1 \cdot 0 - c_2 \cdot 1 + c_3 \\ &= -c_2 + c_3 = 0 \implies c_2 = c_3 \end{aligned}$$

Adding $c_2 = -c_3$ and $c_2 = c_3$, we have

$$\begin{aligned} 2c_2 &= 0 \implies c_2 = 0 \\ &\implies c_3 = 0 \quad \because c_2 = c_3 \\ &\implies c_1 = 0 \quad \because c_1 = -c_3 \\ &\implies c_1 = c_2 = c_3 = 0 \end{aligned}$$

which implies $\cos x$, $\sin x$, and the constant function 1, are linearly **independent** on $\mathcal{I} = (-\pi, \pi)$.

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Linear independence of trigonometric functions**
- .

Linear independence and dependence of functions

Example (Linear independence/dependence)

Consider $\cos^2 x$, $\sin^2 x$, and the constant function 1. Are these three functions **linearly independent** on the interval $\mathcal{I} = (-\pi, \pi)$?

Answer: No, they are linearly dependent.

Solution: We must find constants $c_1, c_2, c_3 \in \mathbb{R}$ (*not all zero*) such that

$$c_1 \cos^2 x + c_2 \sin^2 x + c_3 = 0.$$

One such solution:

$$c_1 = 1, \quad c_2 = 1, \quad c_3 = -1.$$

Linear independence of solutions of linear ODEs

Distinct solutions of a linear ODE are more useful if they are **linearly independent**.

For a *homogeneous linear n^{th} order ODE*, if we have *n linearly independent solutions*, $y_1(x), y_2(x), \dots, y_n(x)$, on an interval $\mathcal{I}$, then we call the set $\{y_1, y_2, \dots, y_n\}$, a ***fundamental set of solutions*** on the interval $\mathcal{I}$.

Theorem (Existence of a Fundamental Set [Trench §9.1])

*Suppose $L[y] = 0$ is **normal on $\mathcal{I}$** . Then there exists a fundamental set of solutions on $\mathcal{I}$.*

This theorem is important because any solution of $L[y] = 0$ can be expressed as a linear combination of the fundamental set...

Linear independence of solutions of linear ODEs

Theorem (General Solution of Homogeneous n^{th} order ODE [Trench 9.1.4])

If $\{y_1, y_2, \dots, y_n\}$ is a *fundamental set of solutions* for the n^{th} order ODE $L[y] = 0$ on an interval $\mathcal{I}$, then the *general solution* of $L[y] = 0$ on the interval $\mathcal{I}$ is

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$$

where c_i , $i = 1, 2, \dots, n$, are arbitrary constants.

- Given this theorem, it is very important to know if a set of solutions is *linearly independent*.
- Checking *linear independence* directly from the *definition* can be cumbersome. Fortunately, there is a convenient criterion based on a determinant that is usually easier to use.

Linear independence of solutions of linear ODEs

Given a set of n functions $\{f_1(x), f_2(x), \dots, f_n(x)\}$ that are at least $n - 1$ times differentiable, the **Wronskian** of the functions is the determinant

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

Example (Wronskian of two functions, x^2 and $\sin x$)

$$W(x^2, \sin x) = \begin{vmatrix} x^2 & \sin x \\ 2x & \cos x \end{vmatrix} = x^2 \cos x - 2x \sin x$$

Linear independence of solutions of linear ODEs

Theorem (Criterion for Linearly Independent Solutions [Trench 9.1.4])

Suppose $L[y] = 0$ is *normal on $\mathcal{I}$* . Consider a set of n solutions $\{y_1, y_2, \dots, y_n\}$ of the n^{th} order homogeneous linear ODE, $L[y] = 0$, on an interval $\mathcal{I}$. The set of n solutions is *linearly independent* on $\mathcal{I}$ **if and only if** the *Wronskian* $W(y_1, y_2, \dots, y_n) \neq 0$ for every $x \in \mathcal{I}$.

- If $y_1, y_2, \dots, y_n$ are solutions of $L[y] = 0$ on the interval $\mathcal{I}$, then *either* $W(y_1, y_2, \dots, y_n) \equiv 0$ on $\mathcal{I}$ (*everywhere zero*) *or* $W(y_1, y_2, \dots, y_n) \neq 0$ on $\mathcal{I}$ (*nowhere zero*).
 - Note: This is not guaranteed for arbitrary functions $y_1, y_2, \dots, y_n$. It is true if $L[y_i] = 0$ on $\mathcal{I}$ for each $i = 1, 2, \dots, n$ (for some n^{th} order linear ordinary differential operator L).
- $\therefore$ To check for linear independence of solutions, it is enough to compute the value of the *Wronskian* at a single point in the interval $\mathcal{I}$.

Linear independence of solutions of linear ODEs

Example (Criterion for Linearly Independent Solutions)

Consider the ODE

$$y'' + y' - 2y = 0 \quad (\heartsuit)$$

Two solutions are $y_1 = e^x$ and $y_2 = e^{-2x}$. Check:

$$y_1 = e^x \implies y_1' = e^x \implies y_1'' = e^x \implies y_1'' + y_1' - 2y_1 = 0$$

$$y_2 = e^{-2x} \implies y_2' = -2e^{-2x} \implies y_2'' = 4e^{-2x} \implies y_2'' + y_2' - 2y_2 = 0$$

Are y_1 and y_2 linearly independent? What is the general solution?

$$\begin{aligned} W(y_1, y_2) &= W(e^x, e^{-2x}) = \begin{vmatrix} e^x & e^{-2x} \\ e^x & -2e^{-2x} \end{vmatrix} = -2e^{-x} - e^{-x} \\ &= -3e^{-x} \neq 0 \text{ for any } x \in \mathbb{R}. \end{aligned}$$

$\therefore \{y_1, y_2\}$ is a *fundamental set of solutions* of $(\heartsuit)$, and the *general solution* of $(\heartsuit)$ is $y = c_1 e^x + c_2 e^{-2x}$, where $c_1, c_2 \in \mathbb{R}$ are arbitrary.

Solutions of nonhomogeneous linear ODEs

Theorem (General Solution of Nonhomogeneous ODE [Trench 9.1.6])

Suppose y_p is any particular solution of the nonhomogeneous n^{th} order linear ODE, $L[y] = g(x)$, on an interval $\mathcal{I}$, and suppose $\{y_1, y_2, \dots, y_n\}$ is a **fundamental set of solutions** for the associated homogeneous equation, $L[y] = 0$, on $\mathcal{I}$. Then the **general solution** of $L[y] = g(x)$ on the interval $\mathcal{I}$ is

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x),$$

where c_i , $i = 1, 2, \dots, n$, are arbitrary constants.

Note: $y_c(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$ is called the **complementary function** for the equation $L[y] = g(x)$. The general solution is $y = y_c + y_p$, where y_p is any solution of $L[y] = g(x)$.

Superposition of solutions of nonhomogeneous linear ODEs

Note: If L is a linear operator for which $L[y_1] = g_1(x)$ and $L[y_2] = g_2(x)$ then $L[y_1 + y_2] = L[y_1] + L[y_2] = g_1(x) + g_2(x)$, so:

Theorem (Superposition Principle for Nonhomogeneous Linear ODEs [Trench 9.1.7])

*Suppose $y_{p_1}, \dots, y_{p_k}$ are k particular solutions (on an interval $\mathcal{I}$) for the **nonhomogeneous** n^{th} order linear ODEs $L[y] = g_i(x)$, where $g_1(x), \dots, g_k(x)$ are continuous functions on $\mathcal{I}$. Then the superposition*

$$y_p(x) = y_{p_1}(x) + y_{p_2}(x) + \cdots + y_{p_k}(x)$$

is a particular solution of

$$L[y] = g_1(x) + g_2(x) + \cdots + g_k(x).$$

Superposition of solutions of nonhomogeneous linear ODEs

Example (Nonhomogeneous superposition)

Consider the linear operator

$$L[y] = y'' + y' + y$$

Note that (writing $L[y] = y + y' + y''$ ∴ it is more natural to compute)

$$L[x] = x + \frac{d}{dx}(x) + \frac{d^2}{dx^2}(x) = x + 1 + 0 = x + 1$$

$$L[x^2] = x^2 + \frac{d}{dx}(x^2) + \frac{d^2}{dx^2}(x^2) = x^2 + 2x + 2$$

$$L[x^3] = x^3 + \frac{d}{dx}(x^3) + \frac{d^2}{dx^2}(x^3) = x^3 + 3x^2 + 6x$$

Consequently, it follows from the [superposition theorem](#) that

$y_p(x) = x + x^2 + x^3$ is a particular solution to

$$\begin{aligned} L[y] &= (x + 1) + (x^2 + 2x + 2) + (x^3 + 3x^2 + 6x) \\ &= x^3 + 4x^2 + 9x + 3 \end{aligned}$$

Practice problems: linear higher-order equations

In [Trench, §9.1](#):

- Verifying a solution of an IVP: Exercise 1(a), (c)
- Checking a [fundamental set](#): Exercise 6(a), (b)
- Computing [Wronskians](#): Exercise 10(a)–(c)
- Uniqueness and the zero solution: Exercise 11
- [Linear dependence](#) of functions: Exercise 16